Phase 8 – Science Features

Project: UKRAA PicoMuon **Scope:** Section 5 of implementation-plan.md (Science plan, §5.1–§5.4), plus two requested extensions: the application of the FFT to muon data to identify cycles, and the use of two or more PicoMuon units as a pointable muon telescope, including the solid-angle mathematics. **Date:** 4 November 2026

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1. Introduction and instrument context

Phase 8 of the implementation plan adds the *science* layer on top of the data pipeline built in Phases 0–7: the `analysis/` tools (`Harmonics.py`, `Forbush.py`, `SolarCycle.py`) and the correction chain in `ComputeCoefficients.py` / `ApplyCorrections.py`. This report explains the physics, the mathematics, and the practical recipes behind each of the four science-plan subsections, with citations to the primary literature.

1.1 What one PicoMuon unit actually is

Each PicoMuon unit contains **two plastic scintillator layers**, each 5 cm × 5 cm × 1 cm, read out by SiPMs, with the layer centres 30 mm apart (20 mm gap between the facing surfaces). The raw data stream confirms this: every event row is tagged `T` (top layer only), `B` (bottom layer only) or `C` (coincidence, both layers within the firmware window). A single unit is therefore already a short, wide-angle **muon telescope** in its own right.

Representative rates measured from the Unit 1 raw files (`examples/RPi4_data/data/raw/2026/…`):

Quantity	Measured value
Top-layer singles rate	≈ 2.4 s^{-1}
Bottom-layer singles rate	≈ 2.3 s^{-1}
Coincidence (C) rate	≈ 880 h^{-1} ≈ 0.24 s^{-1} (≈ 21,000 day^{-1})

The singles rates are roughly ten times the rate expected from muons alone through 25 cm^2 (≈ 0.3–0.4 s^{-1}). The excess is the *soft component* (electrons, positrons and photons from air showers) plus electronic noise near threshold — exactly why the two-layer coincidence is the muon channel [1, 2, 31]. The C rate of ≈ 880 h^{-1} agrees to better than a factor of two with the $\cos^2\theta$-weighted geometric prediction for this geometry (§8.4), which is good agreement given the uncertainty in the low-energy muon intensity, shower multi-particle events, and edge effects in 1 cm-thick scintillator.

1.2 Reference numbers for sea-level muons

From the Particle Data Group review and standard references [1, 2, 32]:

- Vertical integral intensity: I_0 ≈ 70 $m^{-2}\ s^{-1}\ sr^{-1}$ ≈ 0.007 $cm^{-2}\ s^{-1}\ sr^{-1}$ ($p > 1$ GeV/c); ≈ 0.009–0.011 $cm^{-2}\ s^{-1}\ sr^{-1}$ for all momenta.

- Zenith-angle dependence: $I(\theta) \approx I_0 \cdot \cos^n\theta$ with $n \approx 2$ at sea level for $E\mu \approx 3$ GeV (the famous "about 1 muon per cm^2 per minute" through a horizontal surface).
- Mean muon energy at the ground ≈ 4 GeV; muons are produced mostly from pion/kaon decay at 100–200 hPa altitude (≈ 12–16 km).

These numbers anchor every section below.

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2. §5.1 Pressure coefficient β (the barometric effect)

2.1 Physics

The atmosphere above the detector is an absorber. When surface pressure P rises, there is more mass overburden, so more muons are stopped or decay before reaching the ground and the count rate falls. For surface muons the effect is small and negative: $\beta \approx -0.12$ to -0.20 %/hPa [4, 5]. The fitted PicoMuon value of -0.188 %/hPa (finding F11 in the implementation plan) sits comfortably in this range. (Neutron monitors, which respond to the lower-energy nucleonic component, have a much larger coefficient, ≈ -0.7 %/hPa [4, 25].)

2.2 Mathematics

The standard exponential absorption model [4, 5]:

$$N(P) = N_0 \cdot \exp(\beta \cdot (P - P_0)) \qquad \Leftrightarrow \qquad \ln(N/N_0) = \beta \cdot (P - P_0)$$

so the pressure-corrected count is

$$N_P = N \cdot \exp(-\beta \cdot (P - P_0))$$

with P_0 a fixed reference pressure (the long-term station mean, frozen in `config/coefficients.ini`).

β is estimated by **weighted least squares (WLS)** on hourly L2 data. With $x_i = P_i - P_0$, $y_i = \ln(N_i/\bar{N})$ and Poisson weights $w_i = 1/\sigma_i^2$ where $\sigma_i = 1/\sqrt{N_i}$ (the relative error of a count of N_i):

$$\hat{\beta} = \Sigma\, w_i(x_i - \bar{x}_w)(y_i - \bar{y}_w) \,/\, \Sigma\, w_i(x_i - \bar{x}_w)^2$$

$$\sigma(\hat{\beta}) = 1 \,/\, \sqrt{\Sigma\, w_i(x_i - \bar{x}_w)^2}$$

where $\bar{x}_w$, $\bar{y}_w$ are the weighted means. The fit **must include an intercept** (the current gnuplot fit forces the line through zero, which biases β whenever the mean rate drifts).

2.3 Procedure and pitfalls

1. **Remove the primary cosmic-ray variation first.** Forbush decreases and solar modulation move the rate by percent-level amounts that are *correlated with weather systems in time* and will bias β. Either

divide by a pressure-corrected NMDB neutron-monitor rate (scaled) [25], or detrend with a 27-day running mean. Exclude flagged Forbush/GLE days.

2. Fit hourly, not daily: with ≈ 880 counts/hour, 90 days of hourly data give ≈ 2,160 points spanning typically ±30 hPa, so σ(β) ≪ 0.01 %/hPa — far better leverage than daily means.
3. Keep a **rolling 90-day β** for monitoring and a **frozen β** in `config/coefficients.ini` for the correction chain, so that reprocessing history does not change past corrected values.
4. Once a year of data exists, fit pressure and temperature **together** (multiple regression, §3.4), since P and T_eff both vary seasonally and are correlated.

Key references: Dorman's monograph on cosmic rays in the atmosphere [4]; De Mendonça et al. for a modern worked analysis of pressure and temperature effects on muon detectors [5].

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3. §5.2 Temperature coefficient(s)

Two physically distinct temperature effects must be kept apart.

3.1 The atmospheric (true) temperature effect

Muons come from π± and K± decay at the 100–200 hPa level. When the upper atmosphere warms it expands: production moves higher, muons have a longer path and more decay in flight (negative effect); against this, lower air density means pions are more likely to decay to muons before interacting (positive effect). For *surface* detectors, which see mostly low-energy muons, the **negative effect wins**: $\alpha_T \approx -0.1$ to -0.2 %/K, appearing mainly as a seasonal wave [5, 7, 8, 10]. (Deep underground the sign flips: MINOS measures a *positive* ≈ +0.9 %/K because only high-energy muons from decaying pions reach it [9].)

The modern treatment (Barrett et al. [7]; Dmitrieva et al. [8]; Sagisaka [10]) replaces the full temperature profile with one **effective temperature**:

$$T_{eff} = \Sigma_i W_i \cdot T(p_i) / \Sigma_i W_i$$

a weighted average of the temperatures $T(p_i)$ on pressure levels p_i. The rigorous weights W_i come from meson-production theory [7, 8, 9]; a **mass-weighted average** ($W_i = \Delta p_i$, the layer thickness in hPa) is an adequate starting point for a counting experiment of this size, and can be upgraded to the Dmitrieva weights later without changing the pipeline.

The fit, on *pressure-corrected* data:

$$\ln(N_P / N_0) = \alpha_T \cdot (T_{eff} - T_{eff,0})$$

with the same WLS machinery as §2.2. Requirements:

- Temperatures on pressure levels for the station location. The free **Open-Meteo** historical/forecast API provides 1000–30 hPa levels [12]; **ERA5** reanalysis is the research-grade alternative [11]. (`GetDataUpperAir.py` in the plan.)

- At least 6–12 months of data, because the signal is mostly the seasonal wave; with less than a full cycle, α_T and any linear drift are degenerate.
- Once a year exists, fit P and T together:

$$\ln(N/N_0) = \beta \cdot \Delta P + \alpha_T \cdot \Delta T_{eff} + c$$

solved as a standard multiple linear regression (normal equations $\mathbf{X^T W X}\,\hat{\beta} = \mathbf{X^T W y}$), because ΔP and ΔT_eff are seasonally correlated and separate single-variable fits would double-count.

3.2 The instrument temperature effect

The Pico's on-board sensor measures the **detector box**, not the atmosphere. SiPM gain falls with temperature (breakdown voltage rises ≈ 20–60 mV/K), so a fixed discriminator threshold corresponds to a temperature-dependent light level — the counting efficiency drifts with room temperature [31]. Treatment:

- Fit residuals (after P and T_eff correction) against detector temperature over short windows: $\ln(N_{PT}/\langle N_{PT}\rangle) = \alpha_{det} \cdot (T_{det} - T_{det,0})$.
- **Caution:** indoor temperature has a strong daily cycle. Correcting with it can *create or destroy* an apparent solar diurnal variation (§4). The independent check is the **ADC spectrum**: real gain drift shifts the ADC peak position, whereas a genuine rate anisotropy does not. Prefer stabilising the detector temperature over correcting for it.

3.3 The correction chain

Implemented in `ApplyCorrections.py`, applied to L2 hourly files:

$$N_{PT} = N \cdot \exp(-\beta \cdot \Delta P) \cdot \exp(-\alpha_T \cdot \Delta T_{eff}) \; [\cdot \exp(-\alpha_{det} \cdot \Delta T_{det})]$$

All week-to-year plots then carry raw, P-corrected and PT-corrected curves next to the NMDB reference.

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4. §5.3 Daily (solar) variation – harmonic analysis

4.1 Physics

The solar diurnal anisotropy is a genuine ≈ 0.1–0.2 % sinusoidal variation of the cosmic-ray intensity with a maximum typically around 15–18 h local time, caused by the corotation of cosmic rays with the interplanetary magnetic field [17]. It sits on top of (and must be separated from) atmospheric daily cycles and instrument temperature cycles.

4.2 Harmonic fitting mathematics

On PT-corrected hourly data, detrended by a 24-h running mean (or the NMDB ratio), fit the first two harmonics by least squares:

```
y(t) = A₀ + Σ_{k=1,2} [ a_k·cos(2πk·t/24) + b_k·sin(2πk·t/24) ]

amplitude  A_k = √(a_k² + b_k²)        phase  ϕ_k = atan2(b_k, a_k)
```

Least-squares harmonic fitting handles data gaps correctly (unlike a plain FFT, §7.4) and returns amplitudes in physical units directly.

Also produce a **superposed-day plot**: average the fractional deviation in each local-solar-time hour bin over many days, with error bars σ_bin = 1/√(ΣN_bin). This is the most honest visualisation and reveals non-sinusoidal shapes (e.g. a temperature-driven kink).

4.3 Sensitivity

For a Poisson counting experiment, the 1σ uncertainty on a fitted relative sinusoid amplitude is [15]:

```
σ_A ≈ √(2 / N_total)
```

With ≈ 21,000 C events/day:

Duration	N_total	σ_A
1 month	≈ 6.3×10^5	≈ 0.18 %
1 year	≈ 7.7×10^6	≈ 0.05 %

The expected amplitude is 0.1–0.2 %, so **one year gives a ≈ 3σ detection; one month does not**.

4.4 The solar / sidereal / anti-sidereal test

A solar day is 24 h; a sidereal day is 23.934 h. Any *seasonally modulated* solar-time variation leaks symmetrically into the sidereal (24 h − 3.93 min) and **anti-sidereal** (24 h + 3.93 min, a purely artificial frequency) sidebands. The Farley–Storey test [16]: measure all three amplitudes; the anti-sidereal amplitude estimates the seasonal contamination of the sidereal result. A real sidereal (galactic) anisotropy — ≈ 0.05 % at these energies — shows at the sidereal frequency only. Also check that the fitted solar phase does not track the room-temperature cycle (§3.2).

4.5 27-day recurrence

The Sun rotates (synodically) in ≈ 27.3 days, so long-lived active regions and high-speed streams imprint a 27-day recurrence on cosmic rays. Apply the Lomb–Scargle periodogram (§7.5) to *daily* PT-corrected means and look for power near 0.037 day^{-1}; compare with the same analysis on the NMDB station.

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5. §5.4 Solar cycle and Forbush decreases

5.1 Forbush decreases (good prospects)

A Forbush decrease (FD) is a sudden drop in cosmic-ray intensity — hours to onset, ≈ a week to recover — caused by an interplanetary CME and/or high-speed stream sweeping past Earth [20, 21, 22]. Large events give ≈ 1–5 % drops in surface muon rates (5–20 % in neutron monitors). Daily Poisson noise here is ≈ 0.7 % (1/√21,000), so big FDs are clearly visible in 6–24 h averages of one detector.

Detection algorithm (`Forbush.py`):

1. Baseline B(t): median of the previous 27 days of PT-corrected hourly rates (robust to the event itself).
2. Test statistic on a 6-h (and 24-h) moving mean $\bar{R}$:

```
z(t) = ( R̄(t) − B(t) ) / σ        flag if z ≤ −k   (k ≈ 3)
```

with σ the combined Poisson + observed scatter of the baseline window. 3. Cross-check: require a simultaneous decrease in one or more NMDB stations [25] — this kills local/instrumental false positives. 4. Mark candidate times from NASA DONKI ICME/CME lists and Kp/Dst indices [27] (reuse the Magnetometer suite's `space_weather.py`). 5. **Superposed-epoch (Chree) analysis** [23]: align many events on their onset time and average. The mean FD profile emerges from events far too small to see individually. 6. The **muon/neutron drop ratio** δμ/δNM is a crude rigidity-spectrum probe: muon detectors respond to higher median rigidity (tens of GV) than neutron monitors (few GV), and FDs suppress low rigidities more [21, 22].

5.2 The 11-year solar cycle (long-term, harder)

Solar modulation makes cosmic-ray intensity anti-correlate with the sunspot cycle, with a lag of some months [24]. At surface-muon rigidities the full cycle swing is only a **few percent** (vs 15–25 % for high-latitude neutron monitors), so detector stability — SiPM ageing, threshold and temperature drift — is the real limit, not statistics.

Planned approach (`SolarCycle.py`):

- Monthly or 27-day means of the PT-corrected rate.
- Monthly SILSO sunspot number: `https://www.sidc.be/SILSO/INFO/snmtotcsv.php` [26].
- The NMDB monthly rate for the same months [25].
- **Lagged cross-correlation** between rate x and sunspot number y:

```
r(ℓ) = Σ_t (x_t − x̄)(y_{t+ℓ} − ȳ) / ( √Σ(x_t − x̄)² · √Σ(y_t − ȳ)² )
```

Scan ℓ = 0…24 months; cosmic rays should lag sunspots by ≈ 2–12 months depending on cycle phase and polarity [24]. - Track the **muon/NM ratio**: real heliospheric modulation moves both in a known proportion; detector drift moves only the muon channel. The ratio therefore separates instrument drift from real modulation.

Cycle 25 peaked around 2024–25, so the cosmic-ray intensity should *rise* through 2026–2030. With stable detectors this is detectable in ≈ 3–5 years, sooner if several UKRAA stations are combined.

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6. Interlude: why gaps and cadence shape the method choice

Sections 4 and 5 both reduce to finding periodic signals in a noisy, *gapped*, unevenly sampled counting series. The next section treats this properly, because the user-facing deliverable of `Harmonics.py` is a periodogram and its correct interpretation.

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7. Applying the FFT to muon data to identify cycles

7.1 The discrete Fourier transform

For N samples $x_0 \ldots x_{N-1}$ taken at a uniform interval Δt, the DFT is

$$X_k = \sum_{n=0}^{N-1} x_n \cdot \exp(-2\pi i \cdot k \cdot n/N), \qquad k = 0 \ldots N-1$$

computed in O(N log N) by the FFT [18, 19]. The power spectrum is $P_k = |X_k|^2/N$, and bin k corresponds to frequency $f_k = k/(N\cdot\Delta t)$. The relative amplitude of a sinusoid at bin k is $A_k = 2|X_k|/(N\cdot\bar{x})$.

Two hard limits govern everything:

$$\text{Nyquist frequency:} \quad f_{max} = 1/(2\Delta t)$$
$$\text{Frequency resolution:} \quad \Delta f = 1/(N\cdot\Delta t) = 1/T_{obs}$$

For **hourly** muon data: f_{max} = 12 cycles/day — ample for the 1 and 2 cycles/day harmonics. For **daily** means: f_{max} = 0.5 cycles/day — fine for 27-day and annual lines.

7.2 What the resolution limit means physically

The cycles of interest, in cycles per day (cpd):

Signal	Period	Frequency (cpd)
Solar diurnal	24 h	1.00000
Sidereal diurnal	23.934 h	1.00274
Anti-sidereal (artefact test)	24.066 h	0.99726
Semidiurnal	12 h	2.00000
Solar rotation	≈ 27.3 d	0.0366
Seasonal (T_{eff} wave)	365.25 d	0.00274
Solar cycle	≈ 11 yr	0.00025

Note that the solar and sidereal lines are separated by **exactly 1/365.25 cpd** (one extra sidereal cycle per year). Since $\Delta f = 1/T_{obs}$, you need **$T_{obs} \geq 1$ year just to split the solar from the sidereal line**, and several years to measure the sidereal amplitude cleanly. This is the spectral-resolution restatement of the Farley–Storey argument (§4.4) [16].

7.3 Practical FFT recipe for this dataset

1. Work on **PT-corrected hourly fractional deviations**: $x_n = N_PT_{,n}/\langle N_PT\rangle - 1$. Corrections first — otherwise the spectrum is dominated by pressure weather noise (broadband, days-scale) and the seasonal temperature wave.
2. **Detrend** (subtract a 27-day running mean) before looking at diurnal frequencies; long-period power otherwise leaks across the spectrum.
3. **Window** the data (Hann window) to control spectral leakage from the finite record length; quote the window's amplitude correction factor [19].
4. **Zero-pad** (e.g. 4×) for a smoother spectrum display — this interpolates bins but does *not* add resolution.
5. Estimate significance against the Poisson floor: a pure counting series has a flat (white) spectrum with mean power 1/N_total in relative-power units; a line is interesting when it exceeds the local background by the standard false-alarm criterion (§7.5).

7.4 The gap problem → Lomb–Scargle

Real muon records have holes (USB drops, reboots, cron windows). A plain FFT needs uniform sampling; filling gaps with zeros or interpolation injects spurious power at exactly the daily frequencies of interest (gaps are not random — they correlate with maintenance habits). Two correct options:

- **Harmonic least-squares fitting** at the handful of frequencies of interest (§4.2) — gaps simply mean fewer equations.
- The **Lomb–Scargle periodogram** [13, 14, 15], the generalisation of the periodogram to uneven sampling. For data (t_n, x_n) with mean removed:

$$P(\omega) = \frac{1}{2}\cdot\left\{\frac{\left[\sum_n x_n \cos\omega(t_n-\tau)\right]^2}{\sum_n \cos^2\omega(t_n-\tau)} + \frac{\left[\sum_n x_n \sin\omega(t_n-\tau)\right]^2}{\sum_n \sin^2\omega(t_n-\tau)}\right\}$$

$$\text{with}\quad \tan(2\omega\tau) = \frac{\sum_n \sin 2\omega t_n}{\sum_n \cos 2\omega t_n}$$

The time offset τ makes the result invariant to time translation and equivalent to least-squares sine fitting at each frequency [14, 15]. Under the null hypothesis (Gaussian noise), P at one frequency is exponentially distributed, giving the **false-alarm probability** FAP ≈ 1 − (1 − e^(−P))^M for M independent frequencies [14, 15].

Implementation: `astropy.timeseries.LombScargle` (or `scipy.signal.lombscargle`) — both run easily on a Pi for ≤ 10^5 hourly points. VanderPlas [15] is the definitive practical guide (normalisation, aliasing, FAP, floating mean).

7.5 Expected result sequence

Dataset	Method	Expected detection
1 month hourly	superposed day	atmospheric daily cycle only
1 year hourly	harmonic fit / L-S	solar diurnal ≈ 0.1–0.2 % at ≈ 3σ; semidiurnal marginal

Dataset	Method	Expected detection
1 year daily	Lomb–Scargle	27-day recurrence (compare with NMDB)
≥ 2 years hourly	L-S, three-frequency test	sidereal vs anti-sidereal separation
3–5 years monthly	cross-correlation	11-year modulation trend (with NMDB ratio for drift control)

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8. A pointable muon telescope from two (or more) PicoMuon units

8.1 Concept

A single unit's T·B coincidence already defines a telescope with a 30 mm baseline — but a very wide one (acceptance out to ≈ 67° from the axis). By operating **two units one above the other** and demanding a four-fold coincidence $C_1 \cdot C_2$ (or the pair of outermost layers $T_1 \cdot B_2$), the baseline grows from 3 cm to tens of cm, the acceptance cone narrows, and the stack becomes a **directional instrument**: tilt it, and the counting rate measures the muon intensity from that part of the sky.

A key geometric fact: for equal-area, aligned, parallel layers, any straight track through the two *outermost* layers necessarily crosses the inner layers too. So the four-fold geometry is set by the **outermost layer pair**, and the inner layers add redundancy (background rejection and efficiency measurement, §8.6) at no geometric cost.

8.2 Inter-unit coincidence: timing

The two Picos free-run on separate clocks, so coincidences between units must be formed from timestamps. Two regimes:

(a) Offline matching of RPi timestamps (current hardware). Each event row carries `RPiTimestamp` with microsecond *format*, but the real accuracy is limited by USB/serial buffering and host scheduling — realistically ≈ 1–10 ms jitter. Matching rule: pair a C event from Unit 1 with a C event from Unit 2 when $|\Delta t| \le \tau$. The **accidental (chance) coincidence rate** for singles rates R_1, R_2 and full window 2τ is [30]:

$$R_{acc} = 2\,\tau\,R_1\,R_2$$

With $R_1 = R_2 \approx 0.24\ s^{-1}$ (C events) and $\tau = 1$ ms: $R_{acc} \approx 1.2 \times 10^{-4}\ s^{-1} \approx$ **$10\ day^{-1}$** — small compared with the true stacked-telescope rates below (hundreds to thousands per day), so ms timing is already usable *provided the matching is done between C events*, not raw singles (matching raw T/B singles at 2.4 s^{-1} each would give ≈ 1,000 accidentals/day). Verification plot: the histogram of Δt between the two units must show a sharp true-coincidence peak standing on a flat accidental floor; the floor level measures R_{acc} directly and is subtracted.

(b) GPS PPS timing (planned upgrade). A GPS pulse-per-second disciplining both Picos gives ≈ µs absolute timestamps. Then $\tau = 1$ µs ⇒ $R_{acc} \approx 0.01\ day^{-1}$ — negligible — and the same hardware can do **inter-site shower coincidence** with other UKRAA stations (see "Other improvement suggestions" in the implementation plan).

NTP/chrony on the two RPis (or one RPi reading both units, the current two-unit setup) keeps the clocks within a few ms, which is sufficient for regime (a).

8.3 The zenith-angle law the telescope will measure

The telescope's first science target is the angular distribution itself:

$$I(\theta) = I_0 \cdot \cos^n\theta, \qquad n \approx 2 \text{ at sea level [1, 2, 32]}$$

Procedure: fix the separation (e.g. d = 20 cm), point the stack at zenith angles θ_z = 0°, 15°, 30°, 45°, 60°, 75°, 90°, accumulate equal live times, and fit n from

$$\ln R(\theta_z) = \ln R(0) + n \cdot \ln \cos \theta_z$$

Expected systematics: at large θ_z the detector sees partially through buildings (note the overburden in each direction); near θ_z = 90° the rate does not go to zero (n ≈ 2 is an approximation; high-energy muons flatten the distribution [1]).

8.4 Solid angle and geometric factor: the mathematics

Definitions. For a telescope of two parallel plane detectors, the counting rate in an isotropic flux I (particles $cm^{-2}\ s^{-1}\ sr^{-1}$) is R = I·G, where G is the **geometric factor** (also called gathering power, units cm^2 sr) [28]:

$$G = \int_{A_1} \int_{A_2} \cos\theta_1 \cdot \cos\theta_2 / r^2 \; dA_1\, dA_2$$

Here r is the distance between area elements dA_1 and dA_2 and $\theta_1 = \theta_2 = \theta$ is the angle of the connecting line to the (common) normal; for parallel planes separated by d, cos θ = d/r. Sullivan's classic paper [28] is the standard reference for this formalism.

A useful companion quantity is the **effective solid angle**

$$\Omega_{eff} = G / A \qquad \text{(sr, with A the detector area)}$$

i.e. the solid angle that, multiplied by area and intensity, reproduces the rate.

Limit 1 — detectors stacked (d → 0). Every particle through one layer passes the other, so the telescope degenerates into a single horizontal plane exposed to the upper hemisphere:

$$G \to A \cdot \int_{2\pi} \cos\theta \, d\Omega = A \cdot \pi \quad \Rightarrow \quad \Omega_{eff} = \pi \text{ sr} \approx 3.14 \text{ sr}$$

(The *geometric* half-sky is 2π sr; the cos θ projection factor reduces the effective value to π. For the 5 × 5 cm layer: G = 78.5 cm^2 sr.)

Limit 2 — large separation (d ≫ a). Each detector sees the other as a point: every pair of area elements has r ≈ d, cos θ ≈ 1, so

```
G → A₁·A₂ / d²    = A²/d²            ⇒    Ω_eff ≈ A/d²
```

which is the familiar "solid angle subtended by the far detector". For two 25 cm^2 layers at d = 50 cm: G ≈ 0.25 cm^2 sr, Ω_eff ≈ 0.01 sr.

Exact closed form for all separations. The double integral above is π·A_1 times the **radiative view factor** F_{12} between two directly opposed parallel rectangles — a standard result of radiative heat transfer [29]:

```
G = π · A · F₁₂
```

For two identical squares of side a at separation d, writing X = Y = a/d:

```
F₁₂ = 2/(π X Y) · [ ln √( (1+X²)(1+Y²) / (1+X²+Y²) )
                  + X √(1+Y²) · atan( X / √(1+Y²) )
                  + Y √(1+X²) · atan( Y / √(1+X²) )
                  − X·atan X  −  Y·atan Y ]
```

This reproduces both limits above and is trivial to evaluate in Python (≈ 10 lines; it belongs in `analysis/Telescope.py`).

Muon-weighted rate. Because the real flux is not isotropic ($I = I_0 \cos^2\theta$), the predicted vertical-pointing rate is

```
R = I₀ · ∫∫ cos²θ · cosθ₁ cosθ₂ / r² dA₁ dA₂
  = I₀ · ∫∫ d⁴ / r⁶ dA₁ dA₂
```

evaluated numerically (a 10^5-sample Monte Carlo over both planes converges to < 1 %).

8.5 Worked numbers for the PicoMuon geometry

For two 5 cm × 5 cm layers (A = 25 cm^2), as a function of centre-to-centre separation d. G from the exact view-factor formula; R from the $\cos^2$-weighted Monte Carlo with I_0 = 0.007 cm^{-2} s^{-1} sr^{-1} (p > 1 GeV/c — a conservative floor; the all-momentum rate is ≈ 1.3–1.5× higher); FWHM is the full width at half maximum of the on-axis angular response (from the layer-overlap function):

d (cm)	Configuration example	G (cm^2 sr)	Ω_eff (sr)	Predicted R (h^{-1})	R (day^{-1})	FWHM
→ 0	single layer limit	78.5	3.14 (= π)	—	—	—
3	one unit, internal T·B	27.8	1.11	514	12,300	52°
6	two units touching, B_1·T_2	12.2	0.49	265	6,360	27°
9	two units touching, T_1·B_2 ≈	6.45	0.26	150	3,600	19°
10	small gap	5.39	0.22	127	3,040	17°
15		2.59	0.104	63	1,510	11°
20		1.50	0.060	37	890	8.4°

d (cm)	Configuration example	G (cm² sr)	Ω_eff (sr)	Predicted R (h⁻¹)	R (day⁻¹)	FWHM
25		0.97	0.039	24	580	6.7°
30		0.68	0.027	17	410	5.6°
40		0.39	0.0155	9.7	230	4.2°
50		0.25	0.0099	6.2	150	3.4°
53	50 cm gap, outer layers	0.22	0.0088	5.6	130	3.2°

Notes:

- **"Units directly on top of each other":** the usable baselines are the internal 3 cm pair in each unit, the two facing layers ($B_1 \cdot T_2$ — 3 cm plus twice the layer-centre-to-case distance; measure your cases), and the outermost pair $T_1 \cdot B_2$ at ≈ 9 cm + case allowance. The table rows at 6 and 9 cm are placeholders — substitute the measured centre-to-centre values. The formula needs only a, A and d.
- The internal-telescope prediction (514 h^{-1} at $I_0 = 0.007$) brackets the observed ≈ 880 h^{-1} once the all-momentum intensity (×≈1.4) and shower multi-particle coincidences are included — a useful absolute sanity check for `Telescope.py`.
- **Statistics planning:** for a target 5 % rate measurement per pointing you need N = 400 events; at d = 30 cm vertical that is ≈ 1 day, and at $\theta_z = 60°$ ($\cos^2\theta \Rightarrow \times 0.25$) ≈ 4 days. Narrow telescopes buy resolution with observing time — choose d to match the question.
- Angular resolution vs separation is the design trade: d = 10 cm sees a ≈ 17° cone (good for a 5-point zenith scan in a week); d = 50 cm sees ≈ 3–4° (sky-map pixels, but 150 events/day vertical).

8.6 Efficiency calibration with the four-layer stack

With four layers the per-layer efficiency is measured in situ: select tracks defined by three layers and ask how often the fourth fired:

```
ε₄ = N(1·2·3·4) / N(1·2·3)
```

Apply to each layer in turn. This separates real rate changes from SiPM drift — directly supporting the §3.2 and §5.2 drift-control requirements.

8.7 Data to generate

Suggested per-run record (one CSV row per pointing/configuration, `data/telescope/runs.csv`):

```
run_id, utc_start, utc_end, live_s,
d_cm, zenith_deg, azimuth_deg,
n_c1, n_c2, n_coinc, n_accidental_est,
rate_hz, rate_err_hz, pressure_mean_hPa, rate_P_corrected_hz,
tau_ms, notes
```

plus the raw matched-pair file (one row per inter-unit coincidence: both timestamps, Δt, both ADC values) so the analysis can be re-cut later — consistent with the project's preserve-raw-data rule.

Every rate must be **pressure-corrected** (§2) before angles are compared, because a pointing sequence spanning days will otherwise fold weather into the angular distribution.

8.8 Plots to generate

1. **Δt histogram** between Unit 1 and Unit 2 C events: true-coincidence peak on flat accidental floor. The commissioning plot; also measures the effective τ and R_acc.
2. **Rate vs zenith angle** with fitted $I_0\cos^n\theta$ overlay (linear and log–cos axes); residuals panel. The headline science plot.
3. **Polar sky plot**: rate (colour) on a zenith/azimuth polar grid from a pointing campaign; even an 8-azimuth × 4-zenith map shows the $\cos^2$ bowl and any building shadowing.
4. **Rate vs separation d** at fixed vertical pointing, with the G(d) prediction curve of §8.5 overlaid — a direct experimental verification of the solid-angle mathematics.
5. **Angular response curves**: computed G vs tilt for each d (from the Monte Carlo), showing the narrowing FWHM — the "beam pattern" of the telescope.
6. **East–West asymmetry**: rate at $\theta_z = 45°$ East vs 45° West. The geomagnetic field deflects the (positive-excess) muon parents, giving a measurable West excess of order a few percent [2] — a classic experiment feasible at d ≈ 10–15 cm in a few weeks.
7. **ADC spectra in coincidence vs singles** for each layer: the coincidence requirement isolates the muon (Landau-like) peak from the soft-component and noise pedestal — the efficiency/threshold monitoring plot.
8. Long-term: **sidereal-time superposed histogram** from the narrow configuration (ties into §4.4).

8.9 Extensions with more than two units

- **Three stacked units (six layers):** tracking redundancy, better efficiency calibration, and a hardware-free trigger purity cross-check.
- **Two units side-by-side, horizontal baseline:** extensive-air-shower (EAS) coincidences — two muons from the same shower arrive within ≈ 100 ns over tens of metres; with GPS PPS timing this extends to inter-site shower astronomy between UKRAA stations (implementation-plan "Other improvement suggestions").
- **Fixed orthogonal telescopes (one vertical, one at 45°):** continuous differential monitoring — Forbush decreases are anisotropic in their onset phase, and a two-direction muon system sees the loss-cone precursor [21].

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Computed values in §8.5 were generated with the exact view-factor formula and a 4 × 10^5-sample Monte Carlo ($\cos^2\theta$ weighting); scripts to reproduce them belong in `analysis/Telescope.py` (Phase 8 deliverable).